

OLIVER ERNSTER BSc (Hons)

Senior / Principal Python Engineer & Technical Lead

Fleet, Hampshire, UK · +44 (0)7446 038267 · oernster@codecrafter.uk · github.com/oernster · crankthecode.com

PROFILE

Senior Python engineer and technical lead with extensive experience in software engineering, the most recent 13+ years focused on Python backend, API and application systems, across defence, fintech, telecoms, cloud and startup environments. I build and improve production services with a bias toward systems that are reliable, maintainable and straightforward to operate, working hands-on while owning backend architecture and deployment. I focus on structural integrity and architectural clarity, making the decisions that shape a system early and effectively so the right ones compound over time. I work fluently with advanced AI tooling, including RAG, agentic tools and multiple LLMs, applying them through a professionally designed harness and lens so that design decisions and production behaviour remain firmly my own.

CORE SKILLS

- **Python (most recent 13+ yrs):** APIs, backend services, automation and production tooling; strong object-oriented and functional design-pattern experience.
- **Cross-platform:** extensive development across Windows, macOS and multiple Linux distributions.
- **APIs & interfaces (28+ yrs):** designing, consuming and integrating software interfaces in Python, C/C++, JavaScript/TypeScript and other stacks, across backend, application and telecoms systems.
- **Cloud & platform:** AWS (Lambda, CDK, Zappa), Terraform, Docker, CI/CD, deployment automation and production hardening.
- **Frameworks & data:** FastAPI, Flask and Django/DRF; SQL across PostgreSQL, MySQL, MariaDB, SQLite and MS SQL Server, plus NoSQL with Redis, MongoDB and Cassandra.
- **Full-stack:** end-to-end delivery from Python services and APIs through to React/TypeScript and JavaScript front ends.
- **Packaging & distribution:** cross-platform desktop releases, including Linux Flatpaks, signed macOS .dmg images and Windows .exe installers, plus public-facing sites marketing the open-source projects.
- **Modernisation:** legacy migration, Python 2 to 3, dependency rationalisation and API replacement.
- **Leadership:** technical direction, stakeholder alignment and delivery planning; multidisciplinary teams of up to 17.
- **Foundations:** 15 yrs C++ and 4 yrs C, covering multithreaded, multi-process, protocol-oriented, cross-network and interface-driven systems.
- **Go (Golang):** self-directed exploration with several working projects published on GitHub.

PUBLISHED WORK: DECISION ARCHITECTURE

Author of a four-book Decision Architecture series (available on Amazon and referenced at crankthecode.com) on how decision flow, authority boundaries and organisational design shape engineering outcomes, making ownership explicit, reducing decision latency and aligning architecture with the authority needed to operate it.

PORTFOLIO: CRANKTHECODE.COM/PORTFOLIO

Public projects include [Meridian](#) (the reference client for MMSP, an open protocol I designed and specified) and [Clear Budget](#) (a local, day-by-day personal-solvency model), plus [NarrateX](#), [LatencyLab](#) and [Calendrier](#) (Home Assistant integration), spanning cross-OS desktop apps, protocols, web systems and operational tooling. More at crankthecode.com/portfolio.

CAREER HISTORY

Cisco, Senior Python Developer (Consultancy)

2025

- Accelerated platform modernisation and architectural iteration through agentic AI-assisted workflows while preserving production accountability.
 - Brought a legacy Python codebase up to a current, maintainable standard, migrating from Python 2 to 3 and re-platforming services onto FastAPI.
 - Improved release predictability via platform upgrades, dependency rationalisation and idempotent deployment while establishing a modern React/TypeScript interface.
- Tech: Python 2/3, FastAPI, React/TypeScript, CI/CD, AI-assisted workflows.

Harrier, Lead Python Developer (Startup)

2024-2025

- Delivered a serverless Python/FastAPI backend for ingestion, processing and reporting with no standing application infrastructure.
 - Owned architecture, reliability and system behaviour under startup constraints; defined AWS infrastructure in Terraform from first principles.
- Tech: Python, FastAPI, AWS (serverless/Lambda), Terraform, SQL.

- LTI Mindtree, Senior Data Analyst (Consultancy)** 2023
- Replaced a MuleSoft API with a maintainable FastAPI Python 3 platform, reducing operational complexity and cost.
 - Preserved continuity for downstream consumers while improving architecture, coding standards and documentation for handover.
- Tech: Python 3, FastAPI, AWS, SQL, git.
- EPAM, Lead Developer (Consultancy)** 2022
- Prototyped a cloud-based flood-insurance API in a regulated environment, de-risking the architecture across FastAPI, Lambda, PostgreSQL, SQLAlchemy and AWS CDK and giving stakeholders a concrete route for decision-making.
- Tech: Python, FastAPI, AWS Lambda/CDK, PostgreSQL, SQLAlchemy, TypeScript.
- MDRxTech, Senior Developer** 2022
- Delivered Python/FastAPI backend work across Docker, serverless/Lambda and SQL-backed systems for music-technology services.
- Tech: Python, FastAPI, Docker, AWS Lambda, SQL.
- Thales, Senior Developer (Cyber Resilience & Defence Research)** 2021
- Enabled cyber-resilience modelling and visualisation for research users through a multithreaded socket-based backend and Flask front end.
 - Built a targeting web tool with variable range handling that communicated with an encrypted backend API.
- Tech: Python, Flask, React, sockets/multithreading, SQLite.
- Unilink, Senior Developer** 2021
- Delivered reliable Django/DRF software supporting secure family communication under operational and regulatory constraints.
- Tech: Python 3, Django, Django REST Framework, Docker, MySQL.
- Validus Risk Management, Senior Developer (FinTech)** 2020-2021
- Enhanced a regulated wealth-management and risk-analysis platform in line with analytical requirements and business objectives.
- Tech: Python 3, Django/DRF, AWS, React/TypeScript, Docker, SQL, Linux.
- Memset, Senior Developer (Cloud Platform)** 2019-2020
- Strengthened a core Python/Django cloud platform and contributed to platform-level architecture in a pure cloud-engineering team.
- Tech: Python 2/3, Django, Linux, SQL, git.

EARLIER CAREER (1997-2019): SELECTED ROLES

- Fintech (2016-2018):** Bink, GlintPay and Duologi, building Python/Django systems across loyalty, payments and lending startups.
- Media & broadcast (2012-2019):** MPC Advertising, Forbidden and EditShare, using Python/Django, C++, Qt and Objective-C in production media workflows.
- Defence & research:** Hewlett Packard (2012-2014), technical lead / Scrum Master for a 17-person multidisciplinary team; Northrop Grumman (2009-2010) and DERA (1997-1998), covering C/C++, multithreaded and protocol-oriented systems.
- Telecoms:** Anite (1998-2008), progressing from integration and support to development, project management and team lead over 10 years; Semafone (2010-2011) and Malden (2010), using C/C++/Python and SIP/ISDN.
- Security, IoT & other:** Armour Communications (secure comms, 2015-2016), PhotonStar (IoT, 2014), Artfinder (art startup, 2012), Electric Motor Works (EV, 2008-2009) and a Home Office NGO (search/Django, 2010).

EDUCATION

BSc (Hons) Physics, University of Manchester. Final-year project: infra-red reconnaissance satellite design.

INTERESTS

- Open-source software development in my spare time.
- Authoring books: the Decision Architecture series so far, with more planned.
- Self-directed postgraduate-level physics study, for enjoyment rather than a formal qualification.
- 3D printing (8 years): building and configuring multiple printers, now on my tenth, across hardware, electronics, firmware and software, from bed-slingers to CoreXY designs, alongside the printing itself.
- Flight simulation and space MMO gaming with a full HOTAS and rudder pedals setup.
- Foundation amateur radio (ham) licence holder.
- Grade 7 piano (ABRSM).
- Voracious reader of science-fiction, fantasy and popular science.